



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI ARULJOTHI ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

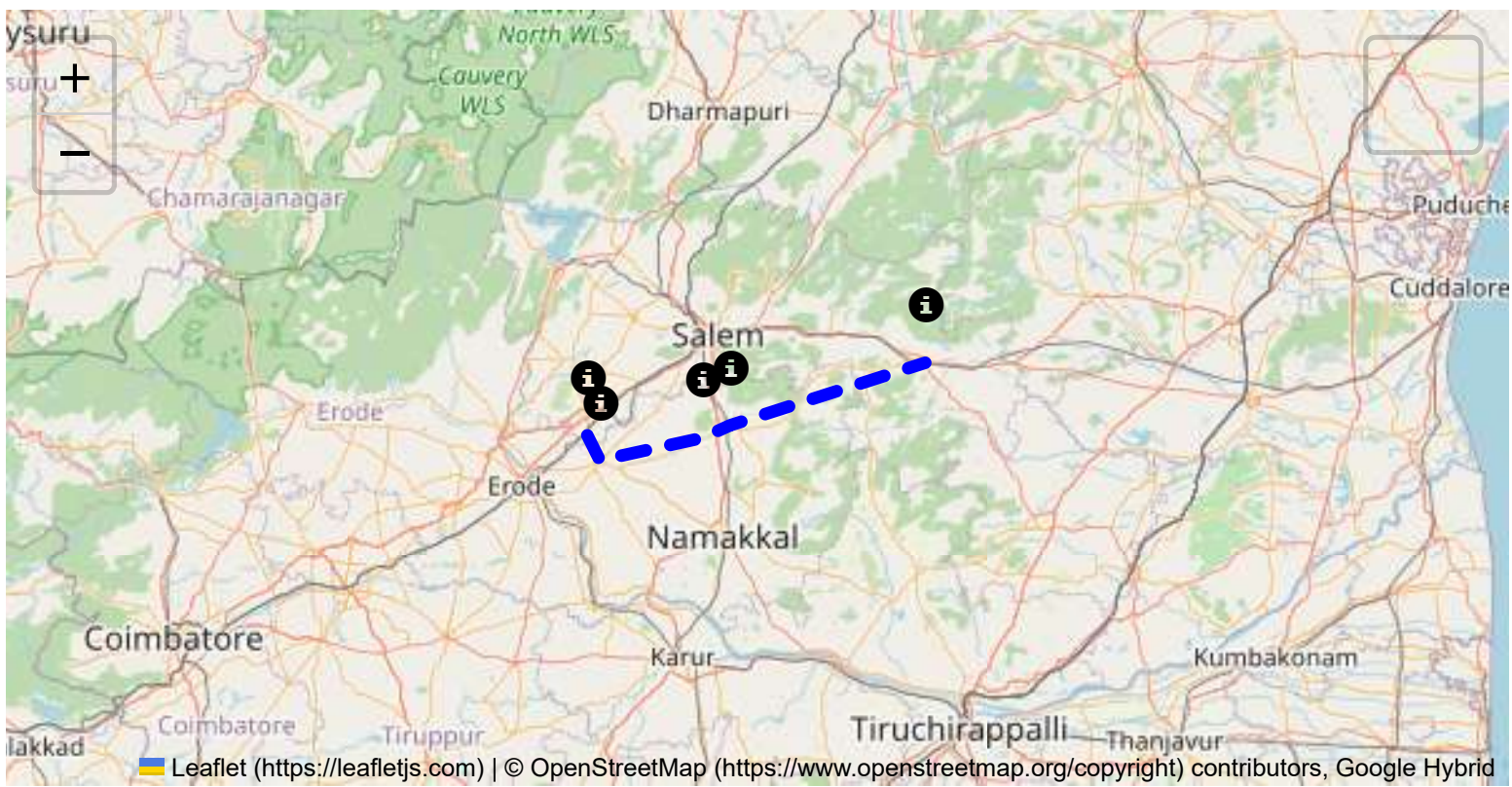
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

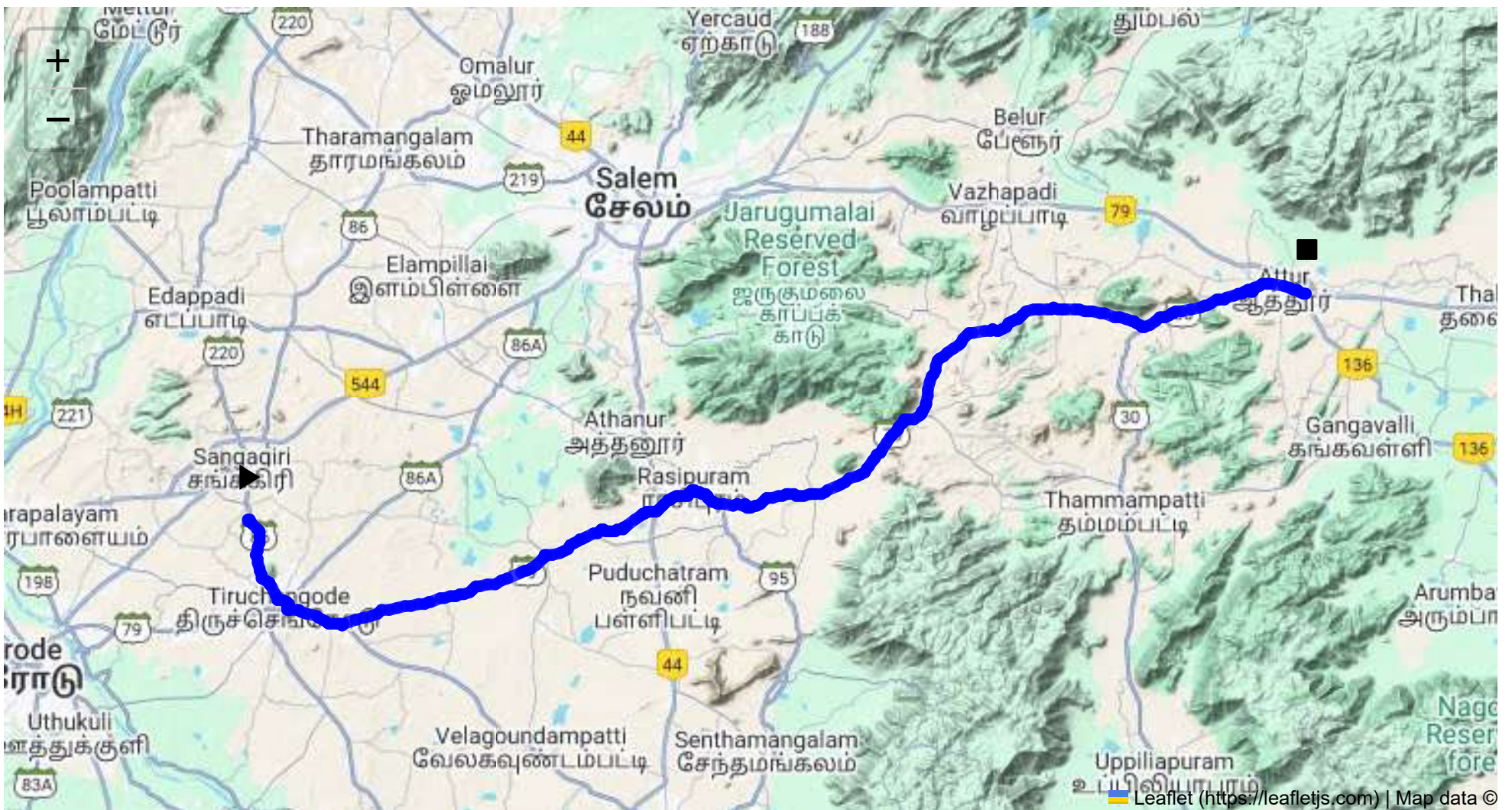
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 99.16 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (11.59377, 78.613091)



Welcome to the Journey Risk Management Study

Route Overview

The route covers approximately 99.16 kilometers, starting from Sangagiri and ending in Attur, Tamil Nadu. It passes through several towns and intersections, including Tiruchengode - Salem Main Rd, Ponkurichi, and critical sections along SH 79. Overall, this route integrates various state highways known for mixed

traffic conditions, particularly involving heavy vehicles due to regional agricultural and industrial activities.

Weather Conditions and Hazards

Tamil Nadu typically experiences a tropical climate. During monsoon seasons (June to September and October to December), heavy rains can lead to water-logged roads, increasing the risk of skidding and accidents. Summers can lead to very high temperatures, which might affect vehicle performance, especially those carrying hazardous materials. Visibility may be compromised on foggy winter mornings, and occasional storms could affect travel.

Traffic Patterns

The route is interspersed with both urban and rural segments:

- **Peak hours:** Morning (8-10 AM) and evening (5-8 PM), especially when local workers and students commute.
- **Congestion:** Likely near towns such as Tiruchengode, Rasipuram, and areas leading into Attur, exacerbated by market days or local events.

Road Quality and Infrastructure

- **State Highways:** These roads are generally well-maintained, but you may encounter narrower sections or patches under repair.
- **Urban stretches:** Expect more frequent speed breakers and pedestrian crossings, requiring careful navigation.

Alternative Routes

In emergencies, consider using NH 44 and diverting through nearby connecting roads to SH 95, which can provide a bypass around congested or blocked areas. Local knowledge can be invaluable for navigating through minor roads if necessary.

Local Regulations on Hazardous Material Transport

- Check for any permits required for transporting hazardous materials.
- Ensure adherence to local loading limits and environmental regulations, particularly in towns that might impose specific restrictions on hazardous cargo.

Historical Incidents

The region has seen accidents involving heavy vehicles due to speeding or poor road conditions. Incidents involving hazardous materials are less common but reinforce the need for strict adherence to safety protocols. Monitoring local news sources or the transport department's advisories could provide real-time risk assessments.

Environmental Considerations

The route doesn't distinctly pass through protected environmental zones but remains sensitive to pollution control. Emission standards must be respected, and all spills should be promptly reported and managed to avoid environmental damage.

Communication Coverage

Major portions of the route have reasonable mobile network coverage. However, isolated segments on less-developed stretches may experience weak signals or dead zones. Carry a fully-charged phone and consider using GPS devices with offline maps for navigation support.

Emergency Response Times

- Urban Areas:** Quick response expected due to proximity to medical facilities and fire stations (within 15-30 minutes).
- Rural Segments:** May have slower emergency response times (up to 45 minutes to an hour) due to distance from major facilities.

Summary of Risk Assessment

Overall, the route poses moderate risk primarily due to traffic congestion points, weather conditions, and occasional road quality issues. Preparation for varied road conditions and adherence to local regulations are imperative. Monitoring for live traffic updates and potential hazards can significantly contribute to a safe journey. Having contingency plans for alternative routes and maintaining careful load management will also enhance transport safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38346, 77.89975	30 KM/Hr
5	Turn	Medium	11.38302, 77.89995	30 KM/Hr
6	Turn	High	11.38188, 77.90257	15 KM/Hr
7	Turn	High	11.37830, 77.90160	15 KM/Hr
8	Turn	Medium	11.36754, 77.93919	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.41361, 78.08073	30 KM/Hr
10	Turn	High	11.41453, 78.08062	15 KM/Hr
11	Turn	Medium	11.41476, 78.08094	30 KM/Hr
12	Turn	Medium	11.43240, 78.12064	30 KM/Hr
13	Turn	Medium	11.43258, 78.12110	30 KM/Hr
14	Turn	Medium	11.44497, 78.15275	30 KM/Hr
15	Turn	Medium	11.45586, 78.18045	30 KM/Hr
16	Turn	High	11.45602, 78.18078	15 KM/Hr
17	Turn	Medium	11.45739, 78.18121	30 KM/Hr
18	Blind Spot	Blind Spot	11.46035, 78.18534	10 KM/Hr
19	Turn	High	11.45904, 78.18556	15 KM/Hr
20	Turn	Medium	11.45132, 78.19979	30 KM/Hr
21	Turn	Medium	11.44994, 78.22138	30 KM/Hr
22	Turn	Medium	11.44919, 78.22291	30 KM/Hr
23	Turn	High	11.44806, 78.22308	15 KM/Hr
24	Turn	Medium	11.44761, 78.22386	30 KM/Hr
25	Turn	Medium	11.49813, 78.32398	30 KM/Hr
26	Turn	Medium	11.50501, 78.33083	30 KM/Hr
27	Turn	Medium	11.50776, 78.33301	30 KM/Hr
28	Turn	Medium	11.50861, 78.33957	30 KM/Hr
29	Turn	Medium	11.51153, 78.34384	30 KM/Hr
30	Turn	Medium	11.51854, 78.34807	30 KM/Hr
31	Turn	Medium	11.54816, 78.35606	30 KM/Hr
32	Turn	Medium	11.56708, 78.37800	30 KM/Hr
33	Turn	Medium	11.58164, 78.41774	30 KM/Hr
34	Turn	Medium	11.58329, 78.44366	30 KM/Hr
35	Turn	Medium	11.57093, 78.49976	30 KM/Hr
36	Turn	Medium	11.57100, 78.50071	30 KM/Hr
37	Turn	Medium	11.57840, 78.51270	30 KM/Hr
38	Turn	Medium	11.59016, 78.55171	30 KM/Hr
39	Turn	Medium	11.59100, 78.55627	30 KM/Hr
40	Turn	Medium	11.59237, 78.55745	30 KM/Hr
41	Turn	Medium	11.59461, 78.56846	30 KM/Hr
42	Turn	High	11.59744, 78.57347	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	Medium	11.59589, 78.57440	30 KM/Hr
44	Turn	High	11.59538, 78.57598	15 KM/Hr
45	Turn	High	11.59892, 78.57708	15 KM/Hr
46	Turn	Medium	11.59982, 78.58078	30 KM/Hr
47	Turn	High	11.60136, 78.58233	15 KM/Hr
48	Turn	High	11.59442, 78.61254	15 KM/Hr
49	Turn	High	11.59395, 78.61240	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
12	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
13	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
14	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
15	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
16	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
17	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
18	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
19	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
20	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
23	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
24	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
25	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
26	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
27	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
29	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium
32	clinic	Ramakrishna Nursing Home	11.5988349, 78.5988615	30 km/h	Medium
33	hospital	Rk Memorial Hospital	11.5975972, 78.5993824	30 km/h	Medium
34	clinic	Sri Vinayaga Physiotherapy Clinic	11.598134, 78.597401	30 km/h	Medium
35	hospital	Sri Andavhar Nursing Home	11.6010351, 78.5974614	30 km/h	Medium
36	clinic	Attur ENT and Dental Care Centre	11.5991073, 78.5962503	30 km/h	Medium
37	hospital	Abhi Eye Hospital	11.5980207, 78.6003698	30 km/h	Medium
38	hospital	Panneerselvam Hospital	11.598067, 78.599939	30 km/h	Medium
39	hospital	Kalyani Memorial Hospital	11.599983, 78.596223	30 km/h	Medium
40	hospital	Sri Dhanvantari Children Hospital	11.5991194, 78.5999856	30 km/h	Medium
41	hospital	Arul Gandhi Physiotherapy Hospital	11.5979315, 78.6003262	30 km/h	Medium
42	hospital	St. Joseph Hospital	11.5990918, 78.6003644	30 km/h	Medium
43	clinic	SS Clinic	11.5975084, 78.5979036	30 km/h	Medium
44	hospital	Gokulam Hospital	11.597356, 78.598403	30 km/h	Medium
45	hospital	Arun Hospital	11.599848, 78.596343	30 km/h	Medium
46	clinic	Bawaa Clinic	11.597934, 78.597143	30 km/h	Medium
47	hospital	Kumar Eye Hospital	11.5978065, 78.5984006	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
48	hospital	Government Hospital Attur	11.59709, 78.601343	30 km/h	Medium
49	hospital	N.S. Kumudham Hospital	11.59626, 78.599512	30 km/h	Medium
50	hospital	Government Hospital	11.597079, 78.600619	30 km/h	Medium
51	clinic	G.V. Homeo Clinic	11.596099, 78.600999	30 km/h	Medium
52	hospital	Lakshitha Hospital	11.5952715, 78.6094785	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
9	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
10	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
11	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
22	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
28	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
30	school	Government Boys Higher Secondary School	11.598743, 78.5885472	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	school	Government Boys Higher Secondary School	11.599149, 78.5888931	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38302, 77.89995



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38188, 77.90257



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37830, 77.90160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36754, 77.93919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41361, 78.08073



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41476, 78.08094



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43240, 78.12064



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43258, 78.12110



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44497, 78.15275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45586, 78.18045



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45602, 78.18078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45739, 78.18121



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46035, 78.18534



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45904, 78.18556



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45132, 78.19979



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44994, 78.22138



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44919, 78.22291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44806, 78.22308



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44761, 78.22386



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49813, 78.32398



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50501, 78.33083



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50776, 78.33301



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50861, 78.33957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51153, 78.34384



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51854, 78.34807



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54816, 78.35606



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56708, 78.37800



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58164, 78.41774



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58329, 78.44366



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57093, 78.49976



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57100, 78.50071



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57840, 78.51270



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59461, 78.56846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59744, 78.57347



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59589, 78.57440



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59538, 78.57598



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59892, 78.57708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59982, 78.58078



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60136, 78.58233

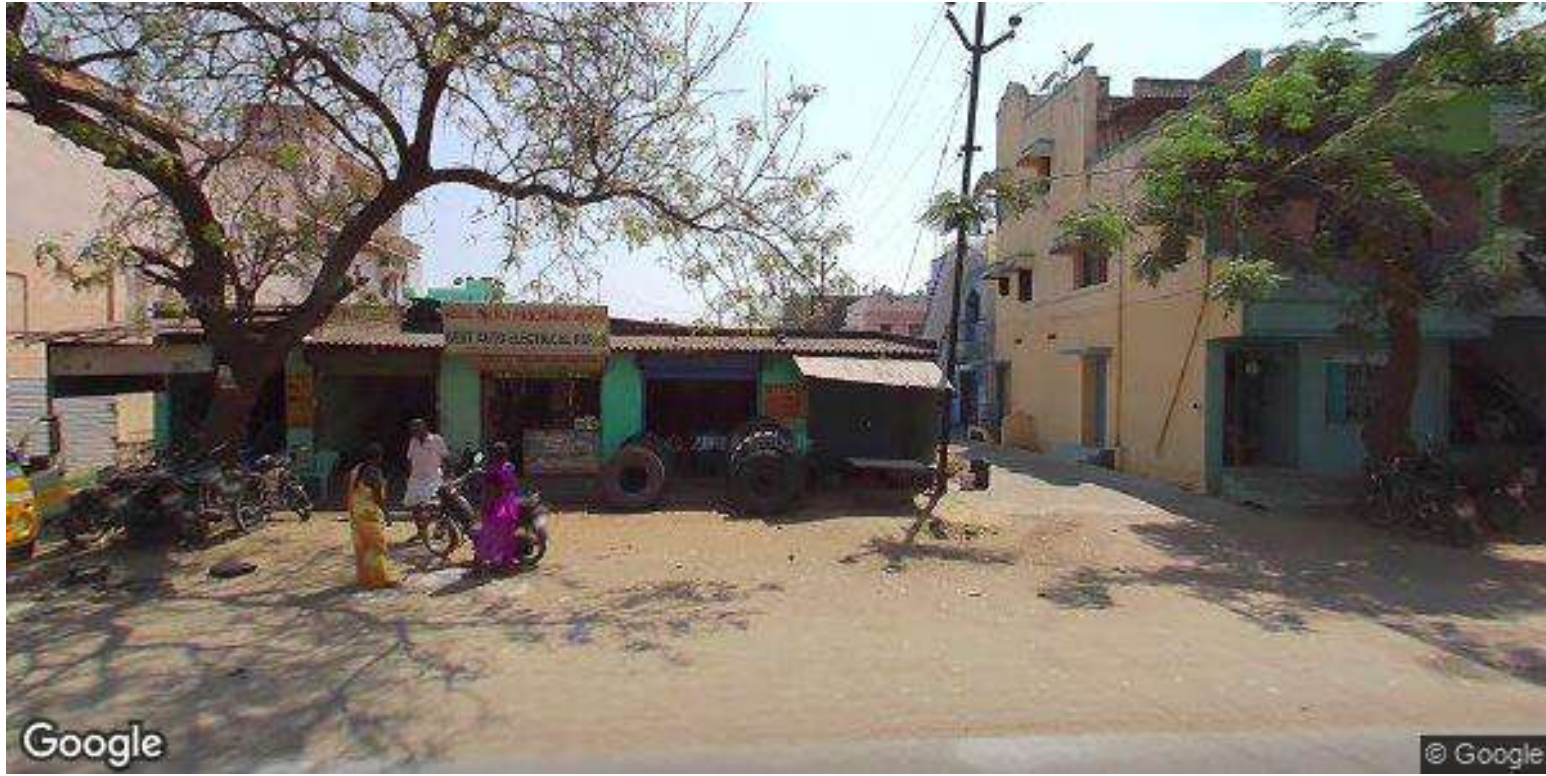


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59442, 78.61254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59395, 78.61240
